The Digestive System

Small intestine
SGL2

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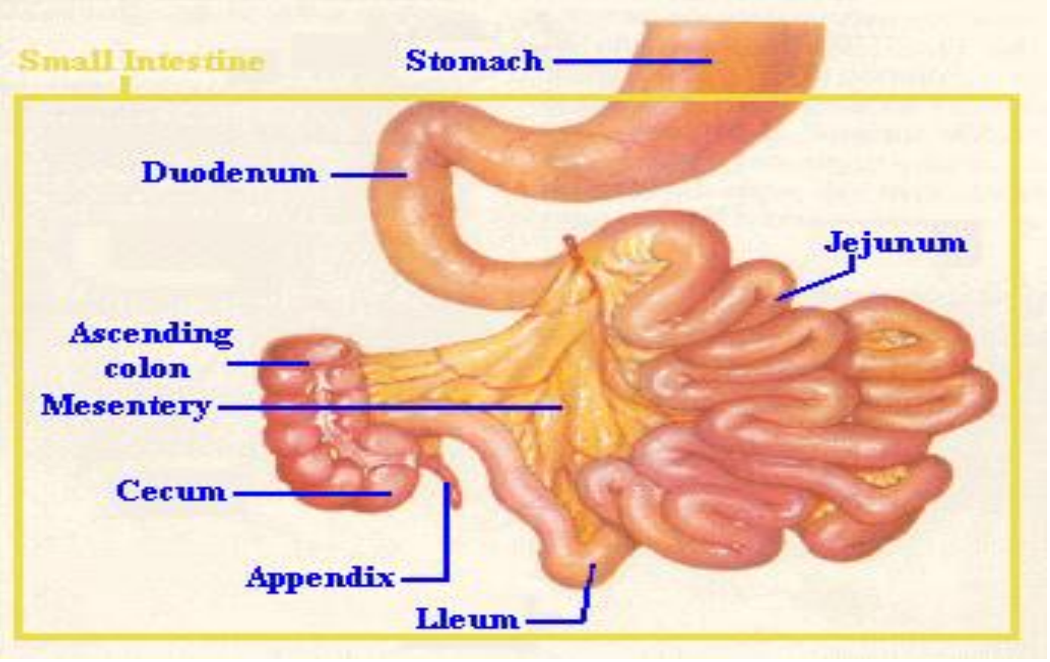
Small Intestine

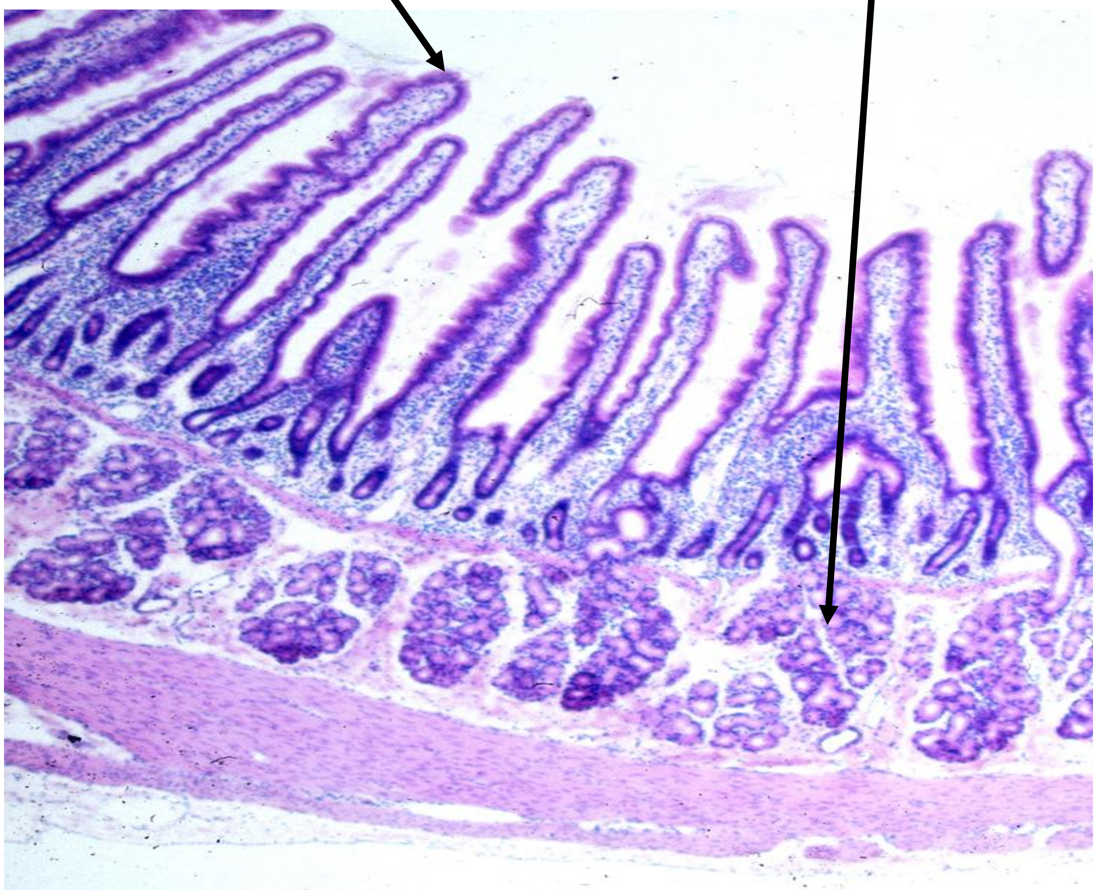
- **The small intestine is relatively long—approximately 5 m this permitting prolonged contact between food & digestive enzyme consist of three regions :**
  - **duodenum** 25cm
  - **jejunum**2.5m
  - **ileum** 3.5m
  - **Its distal end terminates at the ileocecal valve, a sphincter that controls the entry of materials into the large intestine**

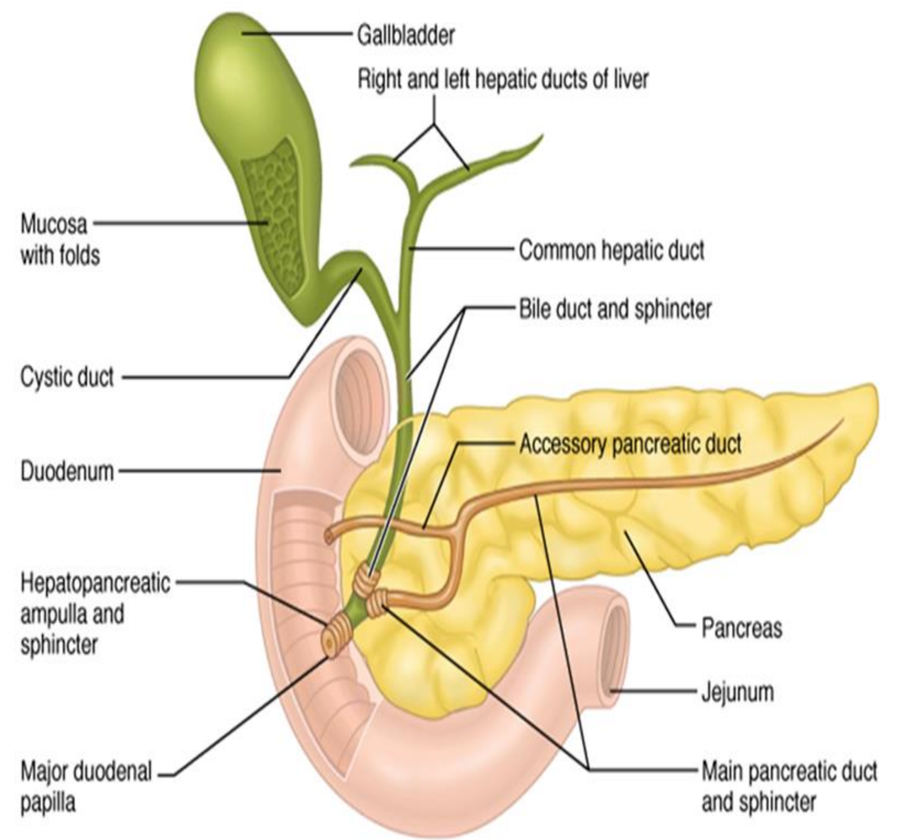
The main functions:

- **absorption**
- **secretion of enzymes**



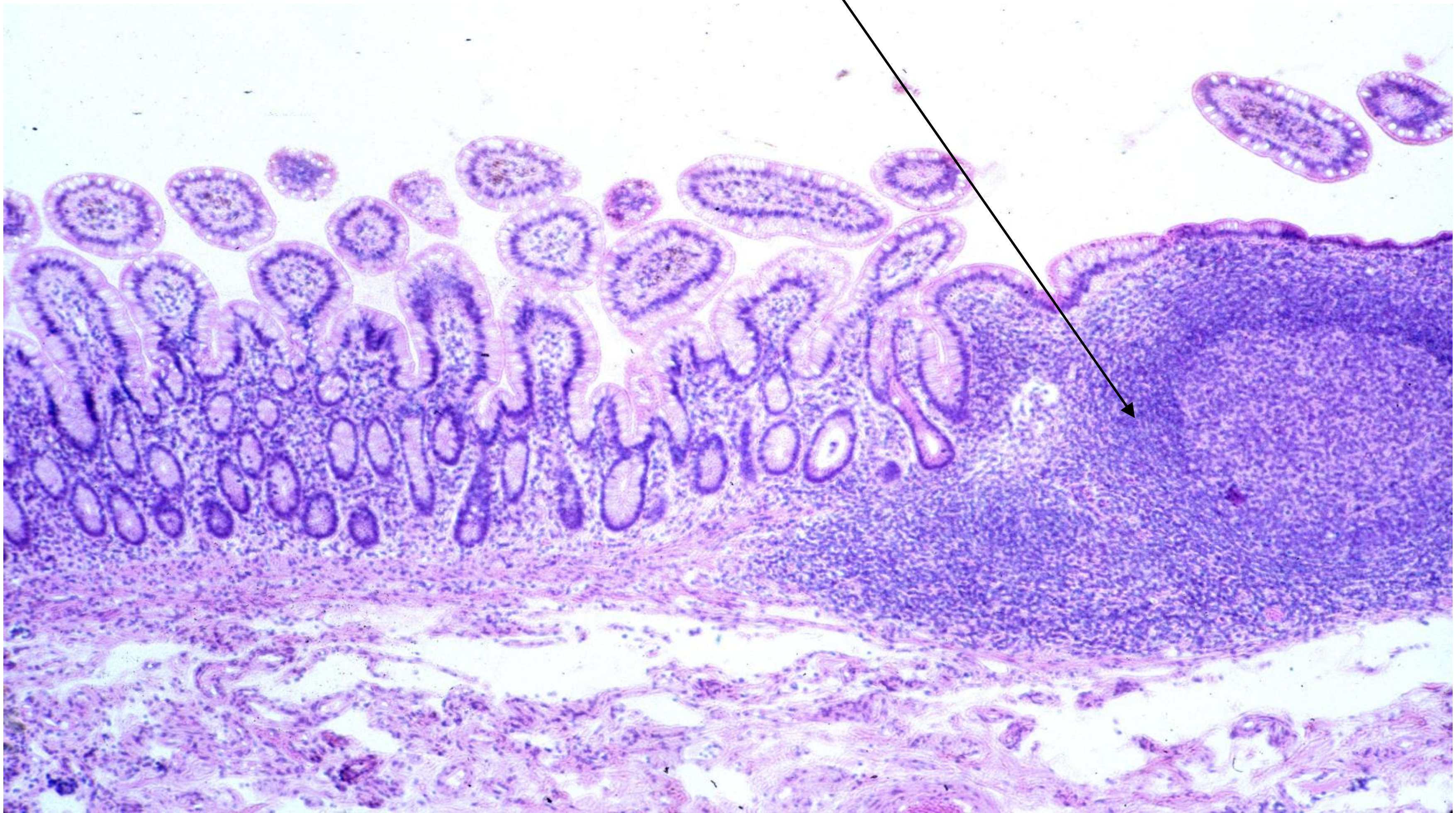
- **Duodenum**
- It has the shape of letter C, bile & pancreatic duct open into it.
- Presence of duodenal(**Brunner's glands**)in the submucosa which secrete alkaline mucus to protect duodenal lining from the acidity of the chyme and bringing the intestinal contents to the optimum pH for pancreatic enzyme action
- has **leaf-shape** villi , few goblet cells , cover by serosa & adventitia





- **Jejunum :extends from the duodenum to the ileum, has long finger-shaped villi, many plicae circulares , intermediate number of goblet cells .**
- **Cover by serosa**

- **Ileum :joins the large intestine at the ileocecal valve , fewer club- shaped villi, abundant goblet cells , Peyer's patches (lymphoid nodules) are found in the lamina propria of the ileum.Cover by serosa**



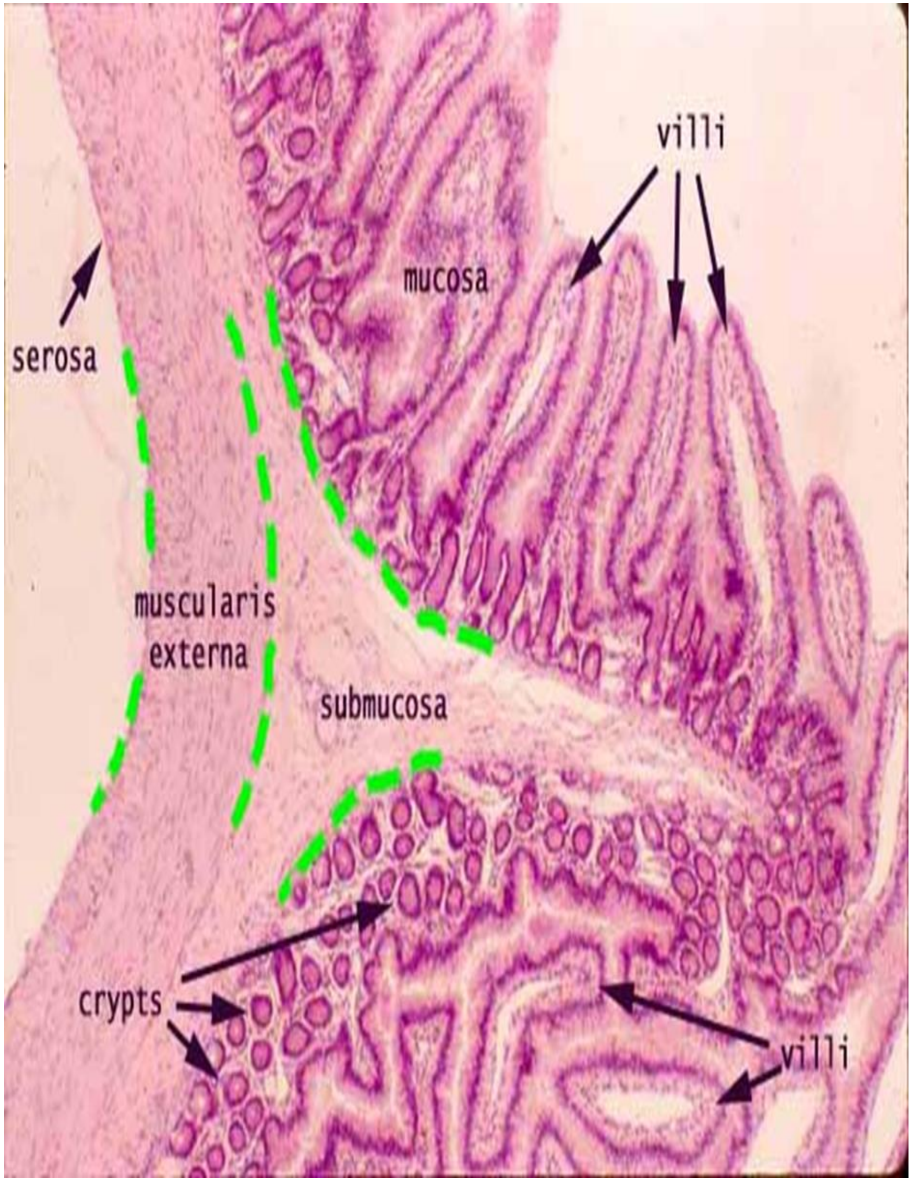
The wall has the same layers as the rest of tract :

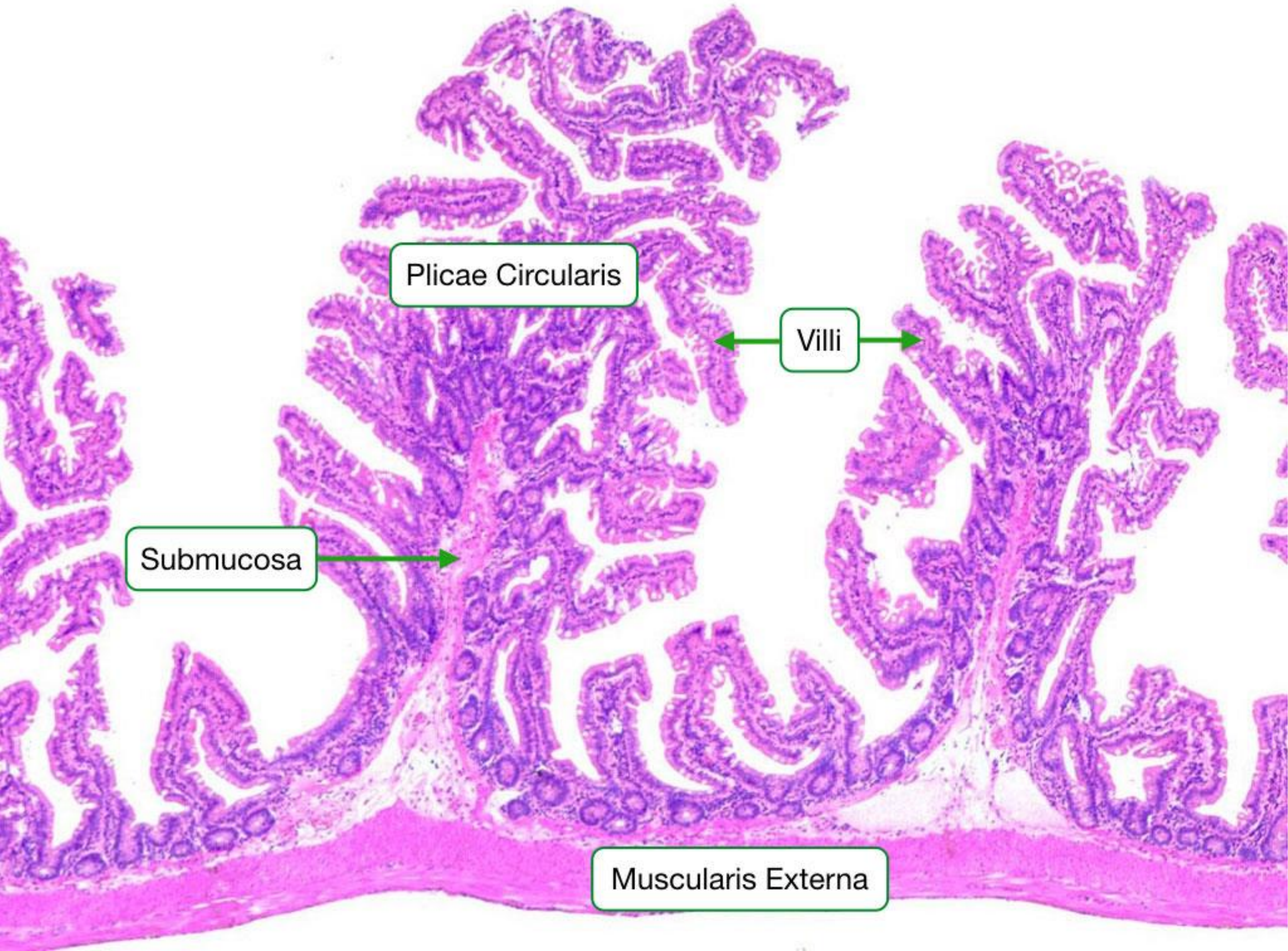
- mucosa
- Submucosa
- Muscularis externa
- Serosa or adventitia

The Mucous membrane

The surface area of the mucous membrane of the small intestine is perform its main function of absorption of food & this is achieved by:
1-The considerable length of the intestine.

- **2-plica circularis: found in jejunum, consist of mucosa and submucosa.The circular folds tend to slow down the passage of contents through the small intestine, thus facilitating absorption.**
- **3-Villi:Are finger-like processes that project from the surface of the mucosa into the lumen.**
- **They consist of a core of reticular tissue covered by surface epithelium.The connective tissue core contains numerous blood capillaries and lymphatic vessel**
- **4-microvilli: surface cytoplasm extension**





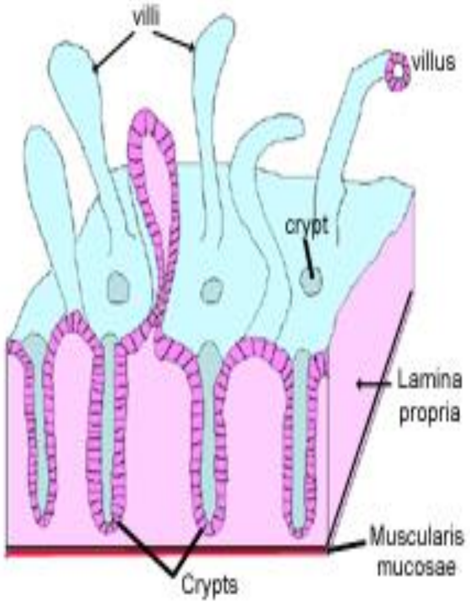
Plicae Circularis
Villi
Submucosa
Muscularis Externa

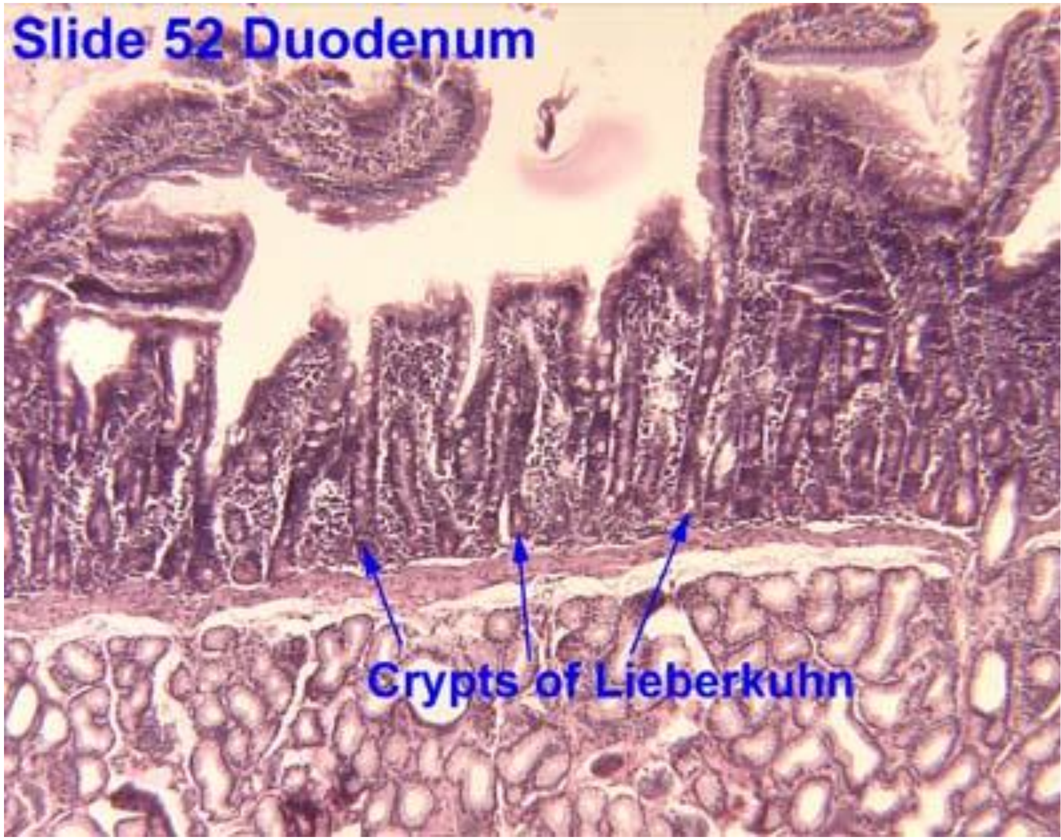
5-Intestinal glands (crypts of lieberkuhn)

Simple tubular glands extend into the lamina propria below the base of villi, they are lined by:

1. -Simple columnar(absorbtive cell)
2. -Goblet cell
3. -Paneth cell
4. -Enteroendocrine cell
5. Stem cells
6. M-cells

Cells of intestinal crypts secrete intestinal juice, There secretion inter lumen through small opening between the villi.

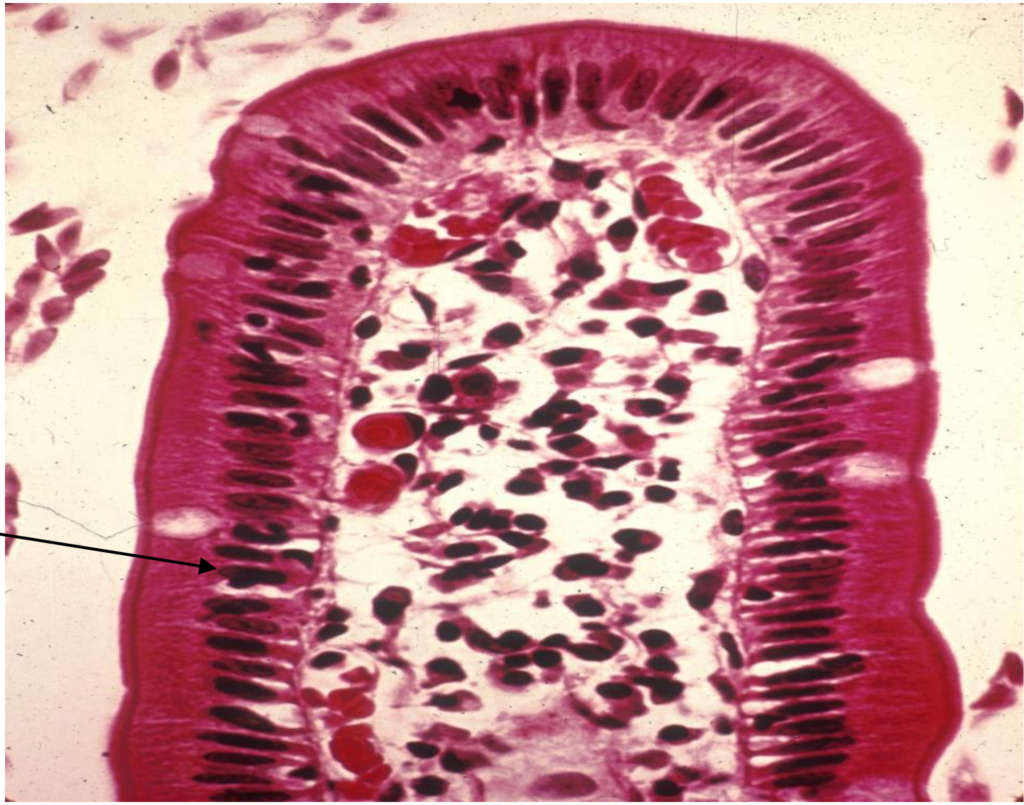


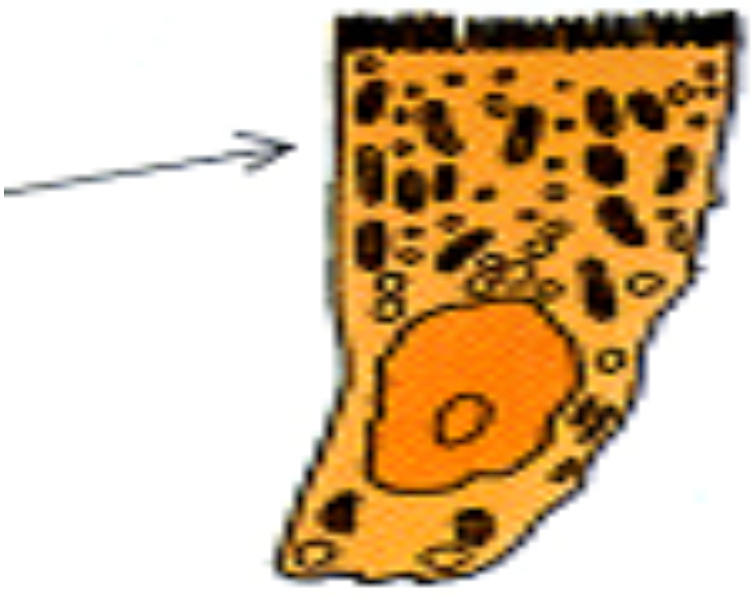


Cells cover the surface & gland of the small intestine :

1-absorptive cells

- Are tall columnar
- Oval basal nucleus
- Has brush border are microvilli
- Each cell has ~ 3,000 microvilli for increased surface area





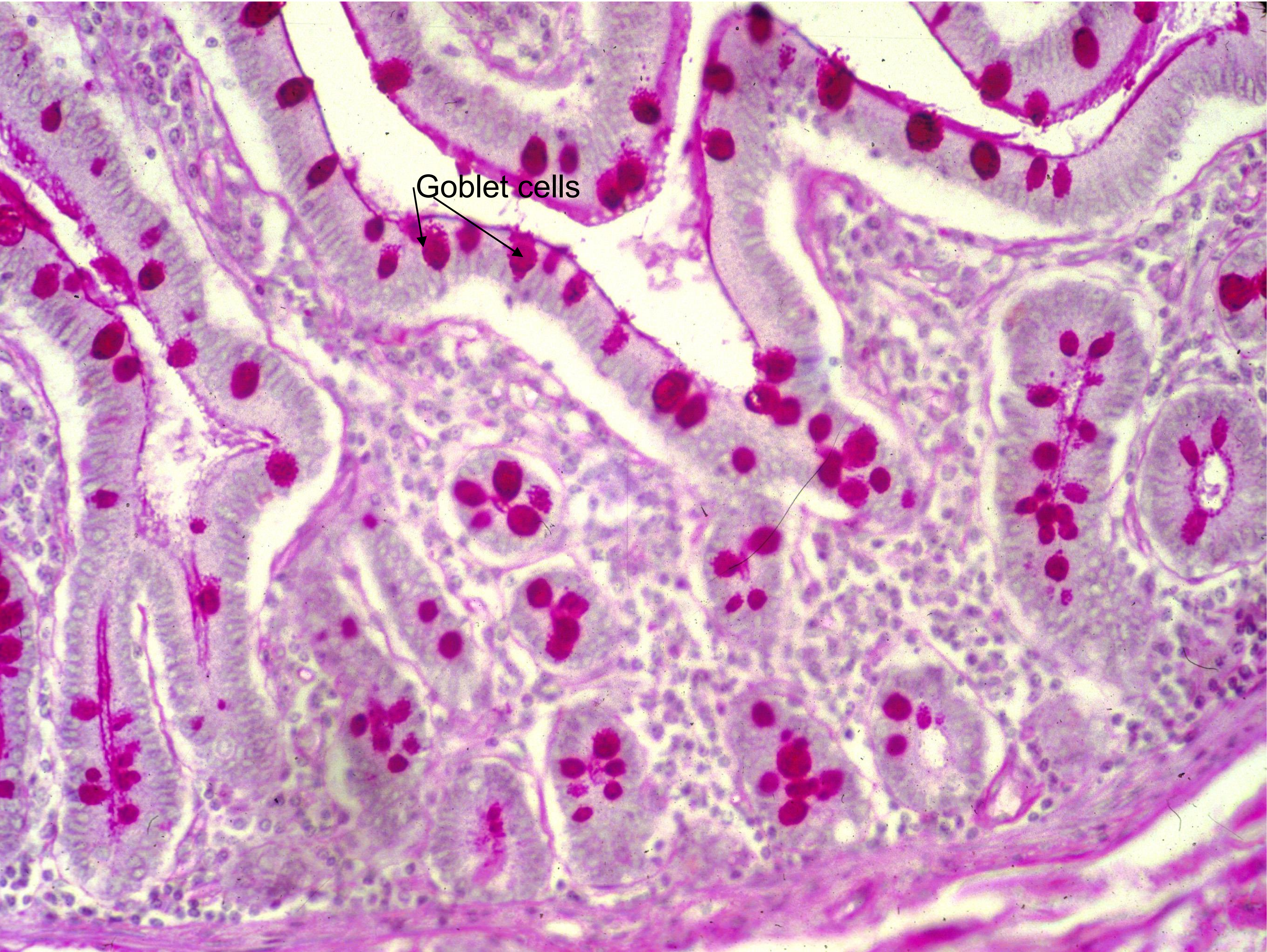
Surface absorptive cell

2-goblet cells: **Are interspersed between absorptive cells**

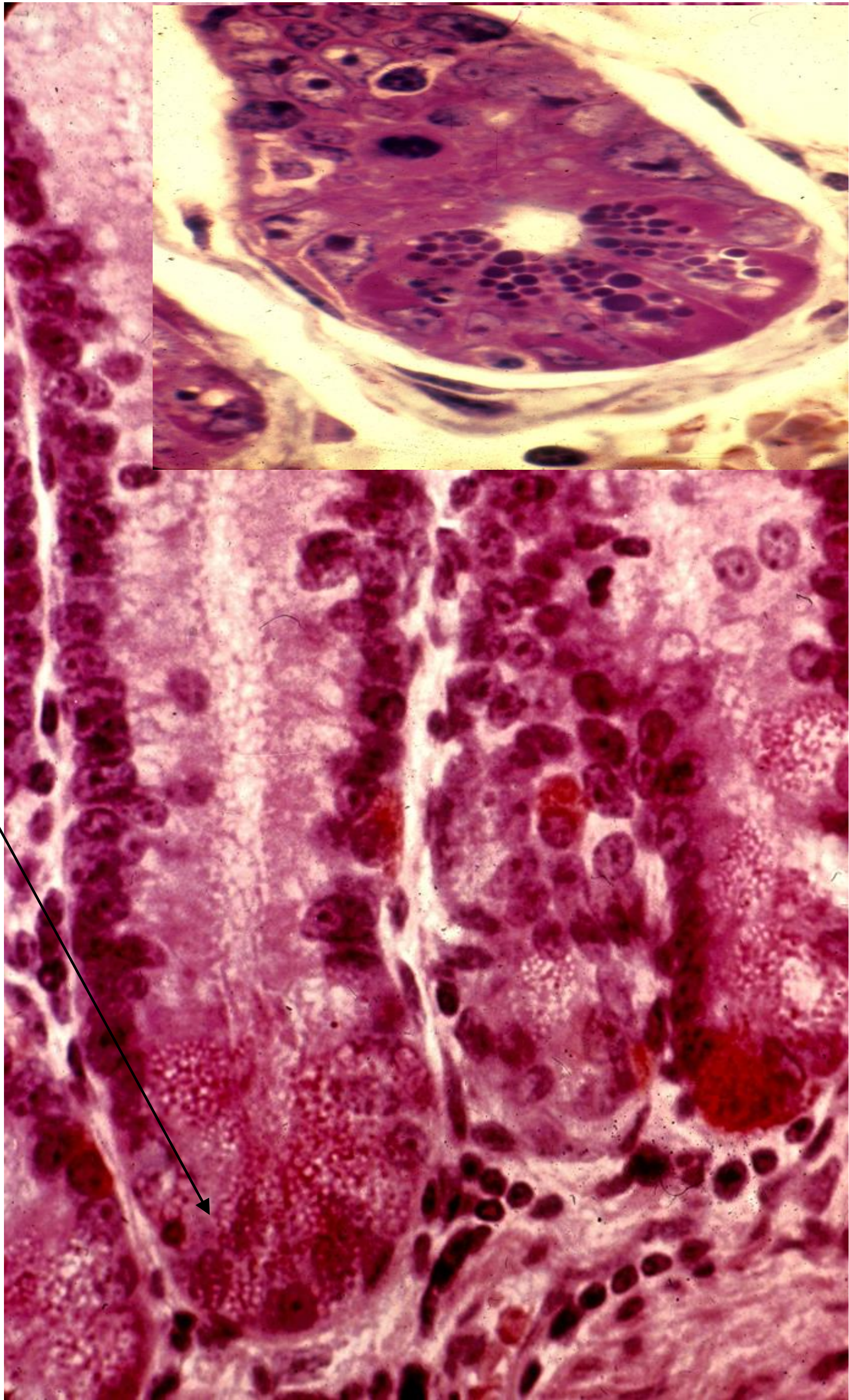
- **Increase in the number as they approach the ileum**
- **Secrete mucus**

for protection

, lubrication

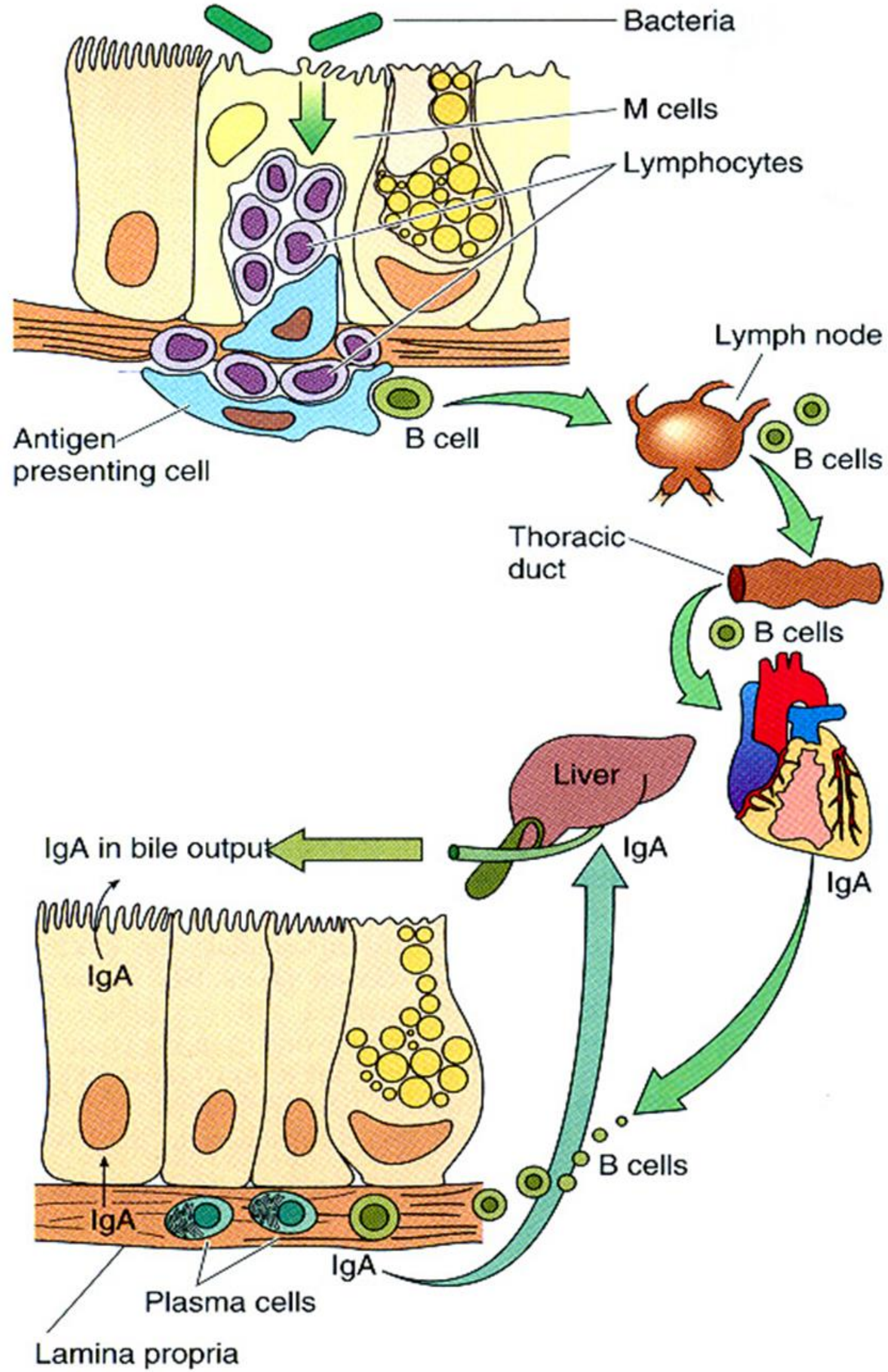

Goblet cells

- 3-paneths cells:
- Located in the basal portion of intestinal glands
- Are exocrine cells , secretary granules in apical cytoplasm
- Basophilic cytoplasm, eosinophilic granules containing lysozyme
- Function : Paneth cell granules release
- lysozyme, phospholipase A2, and hydrophobic peptides called defensins, all of which bind and break down membranes of microorganisms and bacterial cell walls.
- Paneth cells have an important role in regulating the microenvironment of the intestinal crypts.



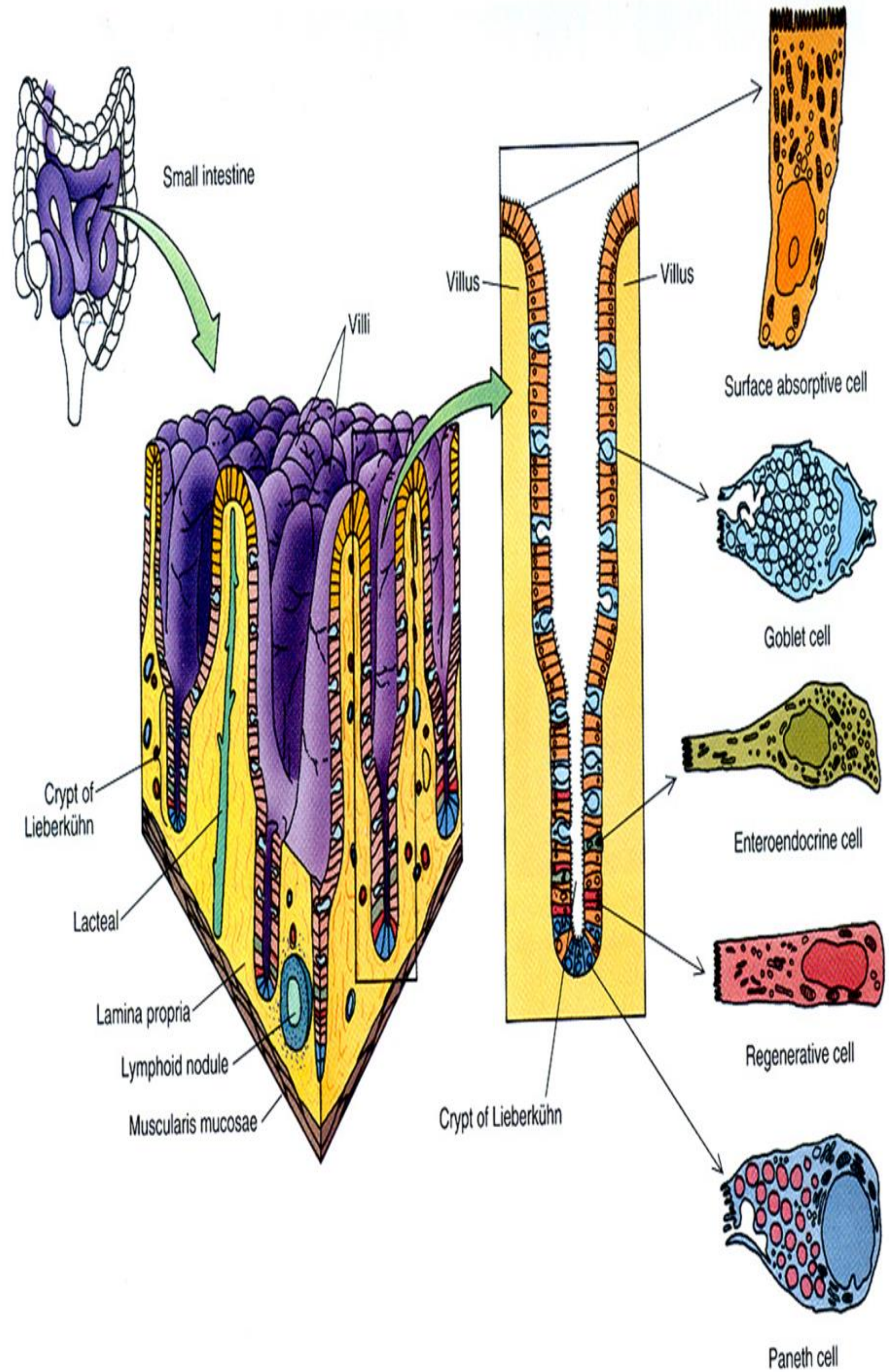
4-M (micro fold) cells:

- **Are specialized epithelial cells, overlying the lymphoid follicles (peyers patches) in ileum**
- **Has discontinuous basement memebrane**
- **These cells can endocytose antigens ,transport to underlying macrophage, lymphocytes cells**
- **Representing protective function in intestine**

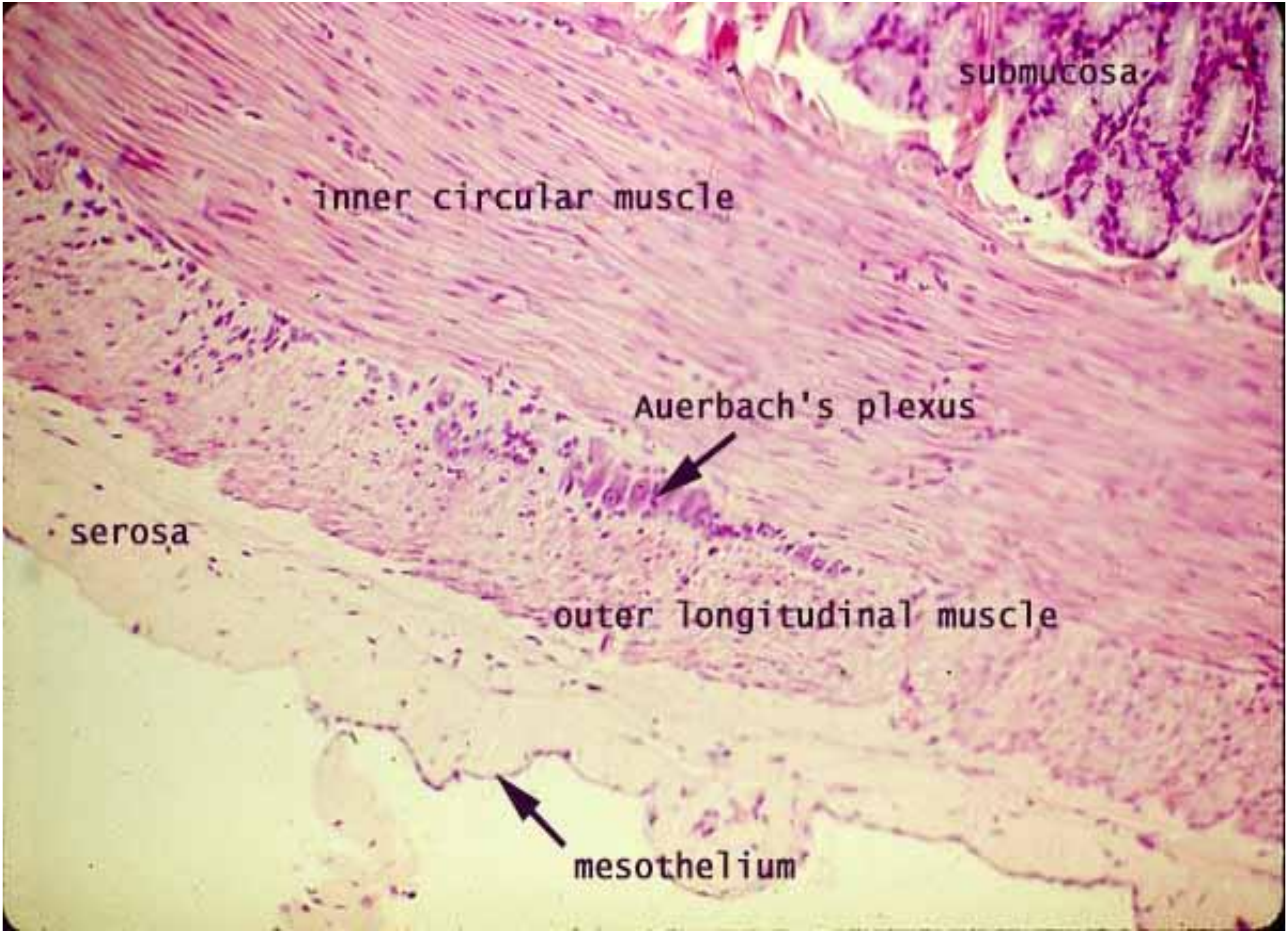


- 5-enteroendocrine cells: Located in lower third of crypt resemble those in stomach APUD cells
- Triangular
- Secrete serotonin, somatostatin, cholecystokinin
- peptides, and secretin
- Regulate intestinal motility and peristalsis, secretions, visceral sensations, and appetite
- 6-stem cells Located in the base of crypt undergo to mitosis, most GI epithelial cells turn over in

  2 - 4 days



- **Lamina propria**: loose c.t. with blood and lymph vessels, nerves, smooth muscle, lymphoid cells and macrophages
- **Muscularis mucosa**: is typical
- **Sub mucosa** : dense c.t. in the duodenum contain Brunner's glands
- **Muscularis externa:** 2 layers inner circular , outer longitudinal
- **Serosa or adventitia**


submucosa
inner circular muscle
Auerbach's plexus
serosa
outer longitudinal muscle
mesothelium

THANKS•

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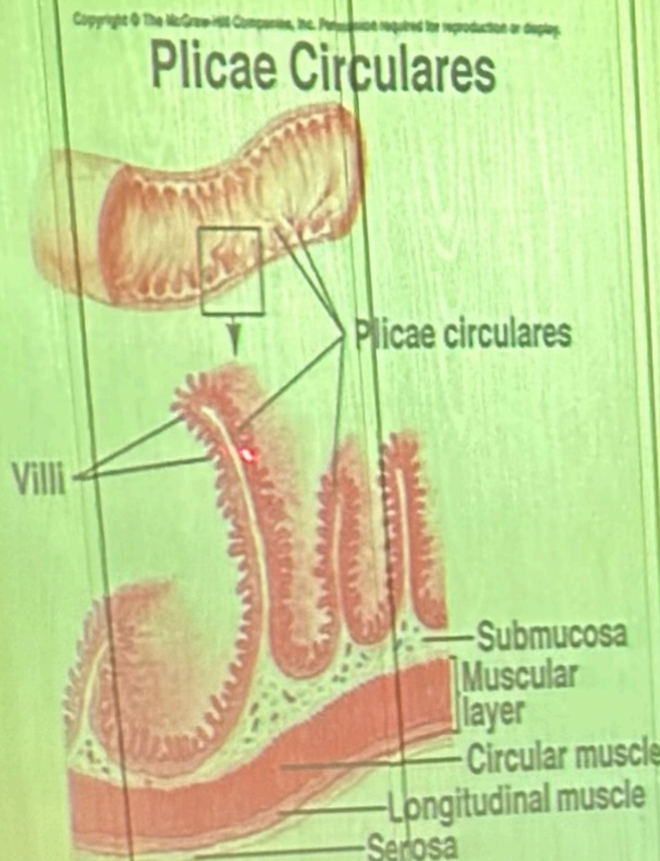
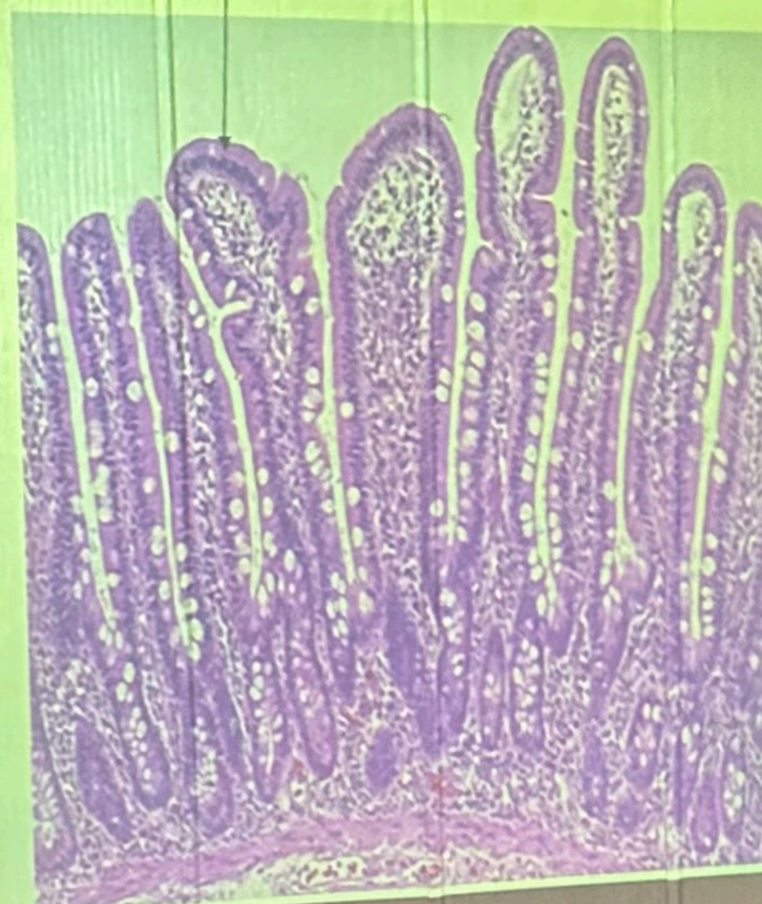
- Cover by serosa





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